PHIL 220g: Introduction to Logic
Practice Midterm

NAME: TA:

Instructions

Two of the given arguments are **propositionally valid** and two are **propositionally invalid**. For each given argument:

1. state it in **premise-conclusion form**
2. **translate** it into **propositional logic**
3. **determine** whether the argument is **valid** or **invalid**:
    - if **valid**, build a **proof** of the conclusion from its premises.
    - if **invalid**, give an **assignment** on which the premises are true and the conclusion false.

OVER PLEASE

A

If you tossed the coin and it landed heads, then it didn't land tails. You tossed the coin, but it landed tails. Therefore, the coin didn't land heads.

PREMISE-CONCLUSION FORM

1.
2.
3.

TRANSLATION INTO PROPOSITIONAL LOGIC

translation key

p:
q:
r:

1.
2.
3.

VALIDITY

Over Please

B

The car will not start unless you turn the ignition and there is gas in the tank. There is gas in the tank. Therefore, the car will start if you turn the ignition.

PREMISE-CONCLUSION FORM

1.
2.
3.

TRANSLATION INTO PROPOSITIONAL LOGIC

translation key

p:
q:
r:

1.
2.
3.

VALIDITY

Over Please

C

If the policy is sound, then inflation will fall. If inflation falls, then the economy will decelerate and unemployment might rise. Therefore, if the policy is sound, then unemployment might rise.

PREMISE-CONCLUSION FORM

1.
2.
3.

TRANSLATION INTO PROPOSITIONAL LOGIC

translation key

p:
q:
r:
s:

1.
2.
3.

OVER PLEASE

D

If your argument is sound, then it is valid and its premises are true. Your argument is not sound. So, if your argument is valid, then its premises are not true.

PREMISE-CONCLUSION FORM

1.
2.
3.

TRANSLATION INTO PROPOSITIONAL LOGIC

translation key

p:
q:
r:

1.
2.
3.

VALIDITY